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# Semisupervised Generative Adversarial Networks With Temporal Convolutions for Human Activity Recognition

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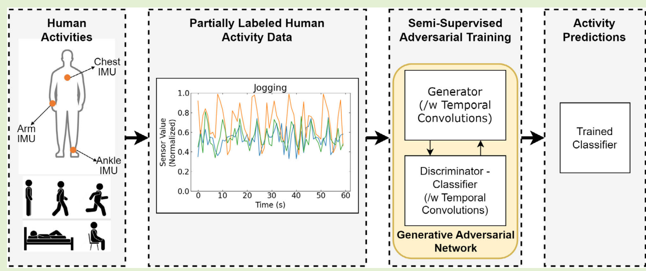
***Abstract*—Many potential applications of human activity recognition (HAR) can be found in health, surveillance, manufacturing, sports, and so on. For instance, HAR can be exploited in ambient-assisted living (AAL) systems to provide users with assistance services intend to improve their well-being, safety, and autonomy. Data annotation in HAR is a complex and time-consuming process that limits the availability of labeled samples. Furthermore, the classification performance of supervised deep neural networks depends on the availability of large, annotated training data. This article proposes an alternative framework for semisupervised generative adversarial networks (GANs) using temporal convolutions for semisupervised action recognition to address several problems related to conventional approaches in the HAR context, such as high dimensionality, scarcity of annotated data, scalability, and robustness. The proposed framework employs a single architecture on different datasets and its effectiveness is tested under four conditions reflecting real-world semisupervised scenarios to investigate the impact of intersubject training, amount of labeled data, number of classes, and inertial measurement unit (IMU) positions on model performance. The evaluations are performed on the Physical Activity Monitoring for Aging People (PAMAP2), opportunity-locomotion, and Laboratory of Images, Signals and Intelligent Systems (LISSI) HAR datasets to evaluate the generalizability of the framework. The results show the proposed framework's high classification performance and generalization ability compared with baseline methods, achieving up to 25% improvement when only a small amount of annotated data is available. A comparison with the previous work using the same datasets has also validated the performance of the framework.**




***Index Terms*— Generative adversarial networks (GANs), human activity recognition (HAR), semisupervised learning (SSL), temporal convolution.**

## I. Introduction

SINCE the last century, along with advances in technology and medicine, life expectancy around the world has risen sharply, which has led to an increase in the ratio of the elderly population to the total population. As a result, more research has focused on improving the quality of life of the aging population in terms of well-being, safety, and autonomy. Ambient-assisted living (AAL) is one such area that aspires to develop solutions for aging well by improving the quality of life of older people and helping them to live and stay active longer through information and communication technologies. A substantial field of research in AAL is human activity recognition (HAR), which seeks to recognize human actions and activities by exploiting sensor data. Some of the prominent use cases of HAR include healthcare, rehabilitation, fitness tracking, activity monitoring, and fall detection. Recently, HAR research has seen significant progress alongside recent breakthroughs in machine learning and deep learning [1].

Nevertheless, there are several challenges when using machine learning algorithms for HAR. One of the major challenges is the dependency on the data collection protocol and the type of activities performed. Indeed, there can be a significant difference between how different participants perform the same activity, which, in turn, influences the intersubject variability within a dataset. Another problem is

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that HAR datasets tend to be smaller than image datasets, as recording and annotating data is time consuming [2]. Therefore, the scarcity of annotated data can decrease the performance of machine learning algorithms [3]. Last but not least, depending on the context of a study, HAR datasets are created using a combination of different sensors. A multichannel dataset can include hundreds of features, and machine learning algorithms may also have difficulty with high dimensionality [4].

There are different approaches to addressing these challenges. Considering that HAR data are frequently collected from different participants, transductive transfer learning techniques can be applied for dealing with inter-subject variability [4]. The issues of high dimensionality and data complexity can be addressed through feature selection, feature extraction, and dimension reduction methods [2], [5]. Traditional HAR systems use manually extracted features with machine learning algorithms, such as, but not limited to, decision trees, Bayesian methods, support vector machines (SVMs), and ensemble methods [2]. However, feature extraction is time consuming, and the extracted features may not be relevant to complex higher level behaviors across various time scales [5]. In this regard, supervised deep neural networks have been demonstrated to be superior to traditional machine learning methods for HAR since they offer automatic feature extraction [5], [6]. However, a significant drawback of deep neural networks is that their performance depends on the availability of labeled data. Finally, when the quantity of labeled data is limited, semisupervised learning (SSL) techniques can be considered as an alternative learning method. [7].

Based on the generative adversarial network (GAN) framework proposed by Goodfellow et al. [8], Salimans et al. [9], and Odena et al. [10] proposed using GANs for semisupervised image classification tasks. Their findings demonstrated that GANs combined with deep architectures can achieve state-of-the-art performance in SSL. In later work, alternative GAN frameworks such as triple GAN [11] and bad GAN [12], further improved their performance.

In its simplest form, a GAN is a deep learning framework consisting of two competing networks: a discriminator and a generator network. These networks are engaged in adversarial training through a min–max game. The generator's task is to generate synthetic samples based on the statistics of the real data, and the discriminator tries to distinguish between the real and generated data. After the initial proposal of GANs, several flaws have been identified, including network instability, nonconvergence [13], and most notably, the mode collapse problem [9], [12], where the generator network constantly fools the discriminator while producing the identical outputs repeatedly.

Although existing studies are still far from fully taking advantage of unsupervised learning methods, the current state of SSL can effectively fill the gap between supervised and unsupervised learning by exploiting unlabeled data. Similar to vanilla GANs [8], semisupervised GANs also consist of a discriminator and a generator network. The main difference is that in semisupervised GANs, the discriminator network simultaneously acts as a classifier, and the goal is to train a classifier by using labeled, unlabeled, and generated data. In theory, by providing both labeled and unlabeled data to the framework, semisupervised GANs can capture domain information from the labeled and unlabeled data and generalize on previously unseen data. According to [9] and [12], another notable difference between semisupervised GANs from vanilla GANs is that the classification performance of semisupervised GANs is inversely correlated with the quality of generated samples since generating samples that are too similar to real data can lead to redundancy. This similarity reduces the information content of the generated samples and lowers their contribution to classifier training.

The primary motivation for using semisupervised GANs over nongenerative SSL methods is to include synthetic data during training. This improves the classifier's performance by allowing it to learn from previously unseen data points and can regularize its performance by generating data from class boundaries [12]. Another benefit of GANs is their scalability due to their flexibility in working with various deep neural network algorithms and datasets of different dimensions, provided that a stable training process is achieved. Moreover, ethical considerations are becoming increasingly important in the field of machine learning, which is also true for the context of HAR. The ethical challenges in this field can arise at various stages, such as data collection, model training, and deployment. Some issues include modeling bias, privacy concerns, algorithm misuse, the black box problem, and sustainability [14], [15]. For instance, in the context of wearable-based HAR, most of the commonly used publicly available datasets have a limited representation of participants of different ages and health conditions, and models trained on such data can be biased. On the other hand, algorithm misuse is also an issue, as one of the main goals of HAR is activity monitoring, which could result in privacy violations. The black box problem is particularly notable in the context of deep learning algorithms, as it limits the reliability and interpretability of neural network models. Similar to computer vision, HAR often requires a large amount of labeled data. This can lead to privacy and sustainability issues due to the time and resources consumed during data labeling. GANs can address issues related to data collection, such as reducing reliance on real-world data, which can help mitigate privacy and sustainability concerns. Moreover, by controlling the generation process, the inherent biases present in datasets can be addressed.

On the other hand, there are some drawbacks to using GANs. One challenge is optimizing the training process and balancing the generator and discriminator networks. GANs are susceptible to mode collapse, a phenomenon where the generator outputs similar or identical samples and the discriminator is unable to differentiate between them. Additionally, the training process can be computationally expensive as multiple forward passes are required during the training. Finally, a significant challenge of using GANs with HAR data is the evaluation of the synthetic data quality. In computer vision, well-established metrics, namely, the inception score and Fréchet inception distance are used to evaluate the quality of generated samples. These metrics are calculated using a pretrained deep neural network, such as the InceptionNet [16]. However, to the best of our knowledge, there is no such evaluation method available for HAR data.

Based on semisupervised GANs in [9], this article proposes an alternative GAN framework with temporal convolutions for end-to-end SSL on high-dimensional HAR data, which is described in detail in Section IV. The performance of the proposed framework is evaluated on three HAR datasets to demonstrate the applicability of a single architecture on different datasets and to reduce the dataset bias in the reported results. Moreover, the effectiveness of the proposed framework is tested under four conditions reflecting real-world SSL scenarios to investigate the impact of cross-subject training, the amount of labeled data, the number of classes, and inertial measurement unit (IMU) positions on model performance. These scenarios not only evaluate the performance of the proposed framework but also provide additional insight into the three datasets under different conditions.

The remainder of this article is organized as follows. Section II discusses the previous literature on supervised learning and SSL approaches for HAR and Section III highlights the contributions of the current work. Section IV presents the proposed framework's architecture and training. The datasets and data preprocessing are discussed in Section V. Section VI presents the training results, and finally, Section VII concludes the findings and provides perspectives for future research.

## II. Related Work

This section presents an overview of the state-of-the-art deep neural network architectures for supervised HAR, followed by a review of the state-of-the-art semisupervised generative models, including adversarial autoencoders (AAEs), variational autoencoders (VAEs), and GANs.

### A. Supervised Learning for HAR

Recent reviews on HAR report that the common supervised deep-learning model choices for HAR are mostly based on CNN, RNN, and hybrid CNN–RNN architectures [1], [17], [18]. In the context of HAR, a typical CNN architecture consists of several convolution layers with or without pooling operations, followed by a feed-forward layer for classification [1], [19]. For RNNs, architectures usually include stacked or bidirectional RNN layers, such as long short-term memory (LSTM) and gated recurrent units (GRUs) [20], [21]. According to [22], while CNNs and RNNs show similar performance in HAR, RNNs are better for modeling temporal dependencies, making them more effective in recognizing short-term activities with natural ordering. In contrast, CNNs excel at feature extraction and are better suited for recognizing long-term repetitive activities. The fact that CNNs excel in extracting spatial features and RNNs are good for learning temporal dependencies led to the development of hybrid models that leverage the strengths of both CNNs and RNNs [1], [5]. Hybrid architectures for HAR appeared in 2016 when [5] introduced the DeepConvLSTM model which consists of four convolutional layers followed by a two-layer LSTM. The classification performance of the architecture has exceeded that of prior CNNs and RNNs.

Following the success of hybrid architectures, many studies employed them to achieve state-of-the-art classification performance in HAR. For instance, Xia et al. [23] proposed an LSTM-CNN architecture, consisting of two-stacked LSTM layers followed by convolutional layers. In another study [24], a hybrid CNN-GRU architecture with multiple branches is proposed. Each branch contains two convolutional layers followed by two GRU units, with different filter sizes to capture different temporal dependencies. [25] proposed an architecture with a CNN in parallel with a bidirectional LSTM, outperforming the previous CNN-GRU architecture. Luwe et al. [26] proposed replacing the LSTM of the DeepConvLSTM [5] with a bidirectional LSTM. Following the previous study, [27] proposed replacing the convolutional layers of the previous work with a residual block of two convolutional layers. Finally, [28] proposed an ensemble learning approach using four different deep learning networks, namely, CNN, stacked LSTM, CNN-LSTM, and ConvLSTM, which improved the classification results compared with single-model approaches.

Autoencoders (AEs) have also found applications in HAR for feature extraction, dimensionality reduction, and noise removal [19]. In [29], a convolutional AE is proposed for unsupervised feature extraction, and the features from the encoder are used in classification. A similar approach is adopted in [30], where the authors propose a pretraining regime with stacked denoising AEs, and the features from the encoders are then used for classification. In another study, [31] proposed an ensemble of AEs, where each AE is trained unsupervised to reconstruct data from a specific class, and the features extracted from the encoder are subsequently used for classification. Additionally, Thakur et al. [32] proposed a hybrid convolutional AE-LSTM architecture. In this architecture, the sensor data are fed through an end-to-end encoder–decoder stack, followed by an LSTM and feed-forward layers for classification.

Attention mechanisms have also been successfully used in CNN, RNN, and hybrid architectures for HAR [33], [34]. For example, in [35], an LSTM coupled with an attention mechanism is proposed and achieves state-of-the-art results. In the case of CNNs, [36] proposed multihead CNN coupled with attention layers. Similarly, [37] proposed an architecture with three separate residual convolutional blocks coupled with an attention mechanism to capture cross-attention between sensor features. In the context of hybrid architectures, [38] proposed an improvement of classification performance over the DeepConvLSTM architecture [5] by including attention layers after the second LSTM layer. Gao et al. [39] proposed the DanHAR architecture, which consists of two residual blocks of convolutional layers coupled with attention layers. The proposed architecture outperformed the DeepConvLSTM [5] significantly. Moreover, both [40] and [41] proposed adding a self-attention layer after the LSTM layers of a CNN-LSTM architecture. Finally, [42] proposed feeding segmented sensor features into parallel CNN-biLSTM branches that are followed by attention blocks.

Some recent studies have employed self-attention-based transformer networks which achieved great success in computer vision tasks [19]. Khaertdinov et al. [43] proposed a fine-tuning scheme using a transformer network on wearable sensor-based HAR data. The parameters of the fine-tuned transformer are frozen and coupled with a feed-forward layer for classification. Shavit and Klein [44] proposed feeding the features extracted by a CNN to a transformer network. Finally, [45] proposed using a transformer model for multilabel sequence-to-sequence activity classification.

Although supervised deep neural networks are proven to be efficient in HAR, a significant drawback is that their performance depends on the availability of labeled data from multiple subjects. This dependence is mainly a result of the nonlinearities, intersubject variability, number of dimensions, and activities present in HAR data [46]. Since labeled data for HAR are difficult to obtain and intersubject variability is high, unlabeled data can potentially be used to improve classification performances. In this regard, SSL techniques can be applied in conjunction with deep neural networks to reduce classification errors related to scarce labeled data [7].

### B. Semisupervised Learning With Generative Models

As described previously, SSL methods can alleviate the issue of scarce annotated data encountered by supervised learning methods by using unlabeled data. Some popular techniques include pseudolabeling, wrapper methods, such as self-training and co-training algorithms, and unsupervised processing steps, such as feature extraction, clustering, or pretraining [7]. Perturbation-based models, manifold modeling, generative models, and active learning methods are also used for SSL [7], [19]. In addition, data augmentation techniques are also commonly applied prior to or during the training of SSL algorithms.

Data augmentation is one of the use cases for generative models in HAR. In the context of deep generative models for data augmentation, [47] proposes a GAN framework to generate synthetic sensor data for training a supervised classifier. The framework is tested on the publicly available HASC2010 corpus dataset [48], collected from triaxial accelerometer recordings of multiple participants performing six different activities. The authors used only three activities in their work, and their results demonstrate that the GANs can be used reliably to generate HAR data. However, the results do not show how GANs behave in a high-dimensional setting or with more activities.

In [49], a semisupervised GAN with a multibranch CNN architecture is proposed. The framework consists of a discriminator, a classifier, and a conditional generator network, with the generator capable of producing conditional samples by taking a noise vector and a specified label as inputs. The framework is evaluated using a publicly available HAR dataset that includes a triaxial accelerometer and gyroscope readings [50]. The performance of the framework is compared with several supervised learning and SSL algorithms, including CNNs, random forests, SVMs, self-training random forests, semisupervised SVMs, and another semisupervised GAN framework. By using 10% of the labeled data, the framework achieved a classification performance of approximately 90% in terms of the F1-score, outperforming the supervised and semisupervised algorithms. However, the proposed framework uses $n$ branches of convolutional layers for $n$ sensors, which makes training the framework computationally expensive when the number of features is high.

Another semisupervised GAN framework is proposed in [4] for semisupervised domain adaptation in the context of the cross-subject domain transfer. The approach aims to transfer the domain of samples from one subject to another using GANs. The proposed framework consists of a discriminator, a classifier, and a generator network with convolutional layers. Additionally, PCA is applied for dimensionality reduction as a preprocessing step. In a cross-subject setting, the framework achieved mean classification performance of 54.5%, 76.1%, and 63.8% in terms of weighted F1-score for the Opportunity-ADL, Physical Activity Monitoring for Aging People (PAMAP2), and Laboratory of Images, Signals and Intelligent Systems (LISSI) datasets, respectively. One of the drawbacks of this approach is that while reducing dimensionality can simplify the training process, it can also result in an information loss. Reported results also show significantly lower classification performance compared with previously supervised classifiers trained on the same datasets [46].

In [51], a VAE framework combined with a CNN classifier is proposed for semisupervised HAR. The classifier network is trained on labeled samples, and the generated unlabeled samples are pseudolabeled. The proposed semisupervised classifier outperforms a supervised classifier by 5.28% when trained on 20% of the labeled data, achieving an accuracy of approximately 65%. Although the approach is promising, the classification performance remains inferior to that of supervised models, such as those reported in [46].

Zhang et al. [52] proposed the adversarial variational embedding (AVAE) framework, which generates exclusive and repeatable latent representations for each input sample. Convolutional layers are used in both encoder–decoder networks and the discriminator network. The framework is evaluated on the PAMAP2 [53] dataset. Based on the results, the AVAE framework is an improvement over the standard VAE model, achieving a classification performance of approximately 78.63% in terms of accuracy, using 20% of the labeled data. As in the previous study on VAEs, the classification performance is inferior to that of supervised classifiers trained on the same dataset [46].

## III. Contributions

The aforementioned studies in Section II uncover several obstacles that hamper the training of an efficient classifier on HAR data, originating from technical and theoretical aspects: 1) high dimensionality due to the spatiotemporal properties of HAR data [46]. 2) Heterogeneity in preprocessing steps, such as dimension reduction, segmentation, and feature selection, can significantly influence classifier performance [4] and lead to significant experimental bias. 3) The scarcity of labeled data, which is common in HAR datasets, limits the performance of supervised classifiers [19]. 4) The use of traditional machine learning methods in conjunction with SSL techniques also leads to poor performance [49]. 5) Preprocessing of multichannel HAR time-series data can result in thousands of features [4] and resulting high/dimensional data are challenging not only for supervised learning algorithms but also for GANs, particularly for the generator network.

To address these challenges, the proposed method implements deep convolutional neural networks to cope with high/dimensional HAR data while ensuring high classification performance. To solve the issues related to the spatiotemporal properties of HAR data, the CNN architectures of the GAN framework are tailored to HAR data by applying convolution, downsampling, and upsampling operations only on the temporal dimension. To the best of our knowledge, this approach has not been applied to semisupervised GANs in previous work. The scarcity of annotated data is addressed by using semisupervised GANs. Compared with the previous work on semisupervised generative models, the proposed framework offers a robust and scalable approach applicable to different datasets with different amounts of features.

## IV. Proposed Method

This section presents the proposed semisupervised GAN framework with temporal convolutions to address some of the open problems identified in previous studies on supervised and semisupervised HAR. Fig. 1 illustrates the architecture and training of the proposed framework.

The semisupervised GAN is implemented by using a CNN architecture for the discriminator/classifier and generator networks. As discussed in Section II, RNN and CNN models exhibit similar classification performance in HAR, although slightly lower than hybrid methods. Depending on the dataset, current state-of-the-art hybrid architectures can outperform simpler CNN architectures by up to 2.9% [24], [25], [26], [28], [41]. In this study, a tradeoff is made between potential small increases in classification performance and a simpler, more stable architecture by only considering CNNs and RNNs. Hammerla et al. [22] mentioned that CNNs outperform RNNs in recognizing prolonged and repetitive activities compared with those with shorter duration and natural ordering. Moreover, [54] demonstrated that CNNs are better suited for high-level activities, while RNNs are better suited for low-level activities. The datasets used in this work consist of repetitive high-level activities (opportunity-locomotion) and medium-level activities (PAMAP2 and LISSI) with predefined ordering, making CNNs better suited. This is confirmed by an evaluation of classification performances of a CNN and a bidirectional LSTM on each dataset in Section VI-B. An additional advantage of CNNs over RNNs is that they are significantly faster for training and classification, and more memory efficient [55].

### A. Temporal Convolution, Pooling, and Upsampling Operations

In computer vision tasks, 2-D convolutional layers with regular/shaped kernels are commonly used for extracting local patterns based on the assumption that adjacent pixels within an image are correlated. However, this assumption may not always be valid for HAR, as samples are collected from different sensors. Therefore, in this work, similar to [46], convolution and pooling operations are only applied to the temporal dimension.

Let $T$ and $S$ denote the temporal and feature axes, respectively. The output feature map $Y$ of a convolution operation on an input map $x \in \mathbb{R}^{T \times S \times C}$ of shape $[T, S, C]$,

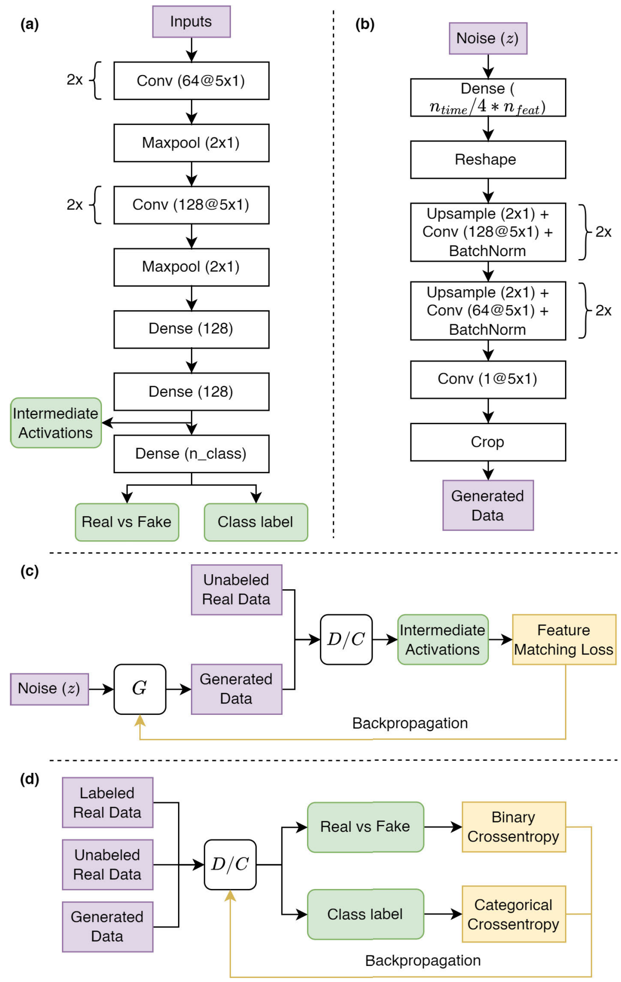


Fig. 1. Illustration of the proposed semisupervised GAN framework. (a) and (b) Architectures of the discriminator/classifier ($D/C$) and generator ($G$) networks, respectively. (c) Training of the $G$ network. (d) Outlines the training of the $D/C$ network.

with a kernel $\mathcal{W}$ of shape $[K, 1, C, F]$, where $C$ and $F$ are the numbers of input and output channels, respectively, is given by

$$Y_{t,s,f} = \sum_{k=0}^{K-1} \sum_{c=0}^{C-1} \mathcal{W}_{k,0,c,f} \cdot x_{t+k,s,c} + \mathbf{b} \quad (1)$$

where $\mathcal{W}_{k,0,c,f}$ is the weight at position $(k, 0, c, f)$, $\mathbf{b}$ is bias, and $t$ and $s$ are the spatial coordinates of the output feature map.

Similarly, max-pooling operations are given by

$$Y_{t,s,f} = \max_{0<p\leq P}(x_{t+p,s,c}) \quad (2)$$

where $p$ stands for the set of values in the temporal axis used for pooling operations. For the generator network, upsampling with nearest neighbor interpolation is chosen over transposed convolutions, as the latter introduces upsampling artifacts [56]. The independence of neighboring data points also impacts nontrainable upscaling operations, and a 2-D upsampling operation on data from multiple sensors will result in an incorrect feature dimension. For an input map of shape $[T, S, C]$, this problem is addressed by upsampling only the temporal axis

$$Y_{t,s,f} = \sum_{m=0}^{T} \sum_{c=0}^{C} x_{2\ m,s,c} \tag{3}$$

where $2m$ stands for the size of the temporal dimension after the upsampling operation.

### B. Semisupervised GAN Framework

The proposed semisupervised GAN framework consists of two separate networks: a discriminator/classifier and a generator network, engaged in adversarial training by optimizing $\min_{\mathcal{G}} \max_{\mathcal{D}/\mathcal{C}} \mathcal{V}(\mathcal{D}/\mathcal{C}, \mathcal{G})$, where $\mathcal{V}$ stands for the value function of the adversarial training. Table I outlines the architecture of the proposed framework.

*1) Discriminator/Classifier Network:* The discriminator/classifier consists of four convolutional layers with two max/pooling operations followed by two fully connected layers before the final output layer. An intermediate output is added after the second fully connected layer to obtain intermediate activations. These intermediate activations are used to calculate the loss of the generator, which will be discussed in more detail in Section IV-C. The discriminator/classifier uses the LeakyReLU activation function for hidden layers.

For a given input $x$, the final layer of the discriminator/classifier $\mathcal{D}/\mathcal{C}(x; \theta^{(\mathcal{D}/\mathcal{C})})$ outputs a $K$-dimensional vector of raw predictions $\{l_1., l_K\}$ and can learn from labeled, unlabeled, and generated samples $x_l$, $x_{\text{ul}}$, $x_{\text{gen}}$. Discrimination between the real and the generated samples is achieved by applying (4) proposed by [9] on the output logits of the classifier model

$$\mathcal{D}(x) = \frac{\mathcal{Z}(x)}{\mathcal{Z}(x)+1}, \quad \text{where } \mathcal{Z}(x) = \sum_{k=1}^{K} \exp[l_k(x)] \tag{4}$$

where $\mathcal{Z}$ corresponds to an intermediate operation. Thus, logits from the discriminator $\mathcal{D}(x)$ are transformed to a value between 0 and 1. For a real sample, the value of $\mathcal{D}(x)$ approaches 1 when a high-confidence prediction is estimated for one of the classes. On the other hand, for a fake sample, this value approaches 0 when all logit values are small. As a result, the discriminator/classifier network can simultaneously yield the realness of a given sample $x$ as well as it's class probability $\mathcal{P}(y|x)$.

*2) Generator Network:* The generator network $\mathcal{G}(z; \theta^{(\mathcal{G})})$ generates synthetic samples that match the distribution of the unlabeled data $p_{\text{data}}(x_{\text{ul}})$ by transforming an input noise vector $z$. The generator architecture consists of a dense layer projecting the input noise $z$, followed by four consecutive blocks of upsampling and convolutional layers. A full-size feature dimension is formed by projecting $z$ before applying temporal operations. The LeakyReLU activation function is used for every hidden layer. Additionally, batch normalization is applied after every convolutional layer, and the final output layer uses the *Tanh* activation function.

The generator's loss is calculated using the feature-matching objective function proposed by [9]. This is achieved by calculating the distance between the intermediate activations of the $D/C$ model on the unlabeled and generated data. The technique is described in more detail in Section IV-C. Previous studies have shown that feature-matching improves SSL performance by reducing the redundancy between generated and real data while generating lower quality but complementary samples [9], [12].

TABLE I

NETWORK ARCHITECTURES OF THE DISCRIMINATOR/CLASSIFIER ($D/C$) AND GENERATOR ($G$) NETWORKS OF THE SEMISUPERVISED GAN FRAMEWORK. *WS*: WINDOW SIZE, *BN*: BATCH NORMALIZATION, AND *I*: INTERMEDIATE OUTPUT. [1], [2], AND [3] INDICATES THE INPUT AND OUTPUT SIZES FOR THE PAMAP2, OPPORTUNITY, AND LISSI DATASETS, RESPECTIVELY

| | Operation | Configuration |
|---|---|---|
| $\mathcal{D}/\mathcal{C}$ | Input | $(90,113,1)^1,(90,40,1)^2,(90,50,1)^3$ |
| | 2×[ Conv2D | [5×1, 64], stride=1, ReLU |
| | MaxPooling | [2×1], stride=(2, 1), ReLU |
| | 2×[ Conv2D | [5×1, 128], stride=1, ReLU |
| | MaxPooling | [2×1], stride=(2, 1), ReLU |
| | 2×[ Dense | 128, ReLU, *I* |
| | Dense | $12^1,5^2,8^3$ |
| $\mathcal{G}$ | Input | (100, 1) |
| | Dense | (ws/16 × n_features, ×128), LReLU |
| | Reshape | (ws/16, n_features, 128), LReLU |
| | 2×[ UpSampling | [2×1] |
| | Conv2D | [5×1, 128], stride=1, BN, LReLU |
| | 2×[ UpSampling | [2×1] |
| | Conv2D | [5×1, 64], stride=1, BN, LReLU |
| | Cropping2D | (ws × n_features × 64) |
| | Conv2D | [5×1, 1], stride=1, BN, tanh |

### C. Semisupervised GAN Training Algorithm

In the discriminator/classifier network, the categorical cross-entropy loss function is used for the multiclass classification of labeled samples, and the binary cross-entropy loss function is used for discriminating between unlabeled samples and generated samples. The total loss of the discriminator/classifier $\mathcal{D}/\mathcal{C}(x)$ is computed by: $l_{\mathcal{D}/\mathcal{C}} = l_{\text{supervised}} + l_{\text{unsupervised}_{\text{ul}}} + l_{\text{unsupervised}_{\text{gen}}}$, where $x_l$, $x_{\text{ul}}$, and $x_{\text{gen}}$ denote labeled, unlabeled, and generated samples, respectively. The total loss of discriminator/classifier is expressed as

$$l_{\mathcal{D}/\mathcal{C}} = \sum_{c=1}^{c} y_l^{(i)} \log\left(\mathcal{C}\left(x_l^{(i)}\right)\right) + \log\left(\mathcal{D}\left(x_{\text{ul}}^{(i)}\right)\right) + \log\left(1 - \mathcal{D}\left(\mathcal{G}\left(z^{(i)}\right)\right)\right). \tag{5}$$

The loss function of the generator is calculated using feature matching in terms of the mean absolute error between the intermediate activations on $x_{\text{gen}}$ and $x_{\text{ul}}$. Let $\bar{x}_{\text{ul}}$ and $\bar{x}_{\text{gen}}$ denote the vectors of means of the rows of vectors produced by the intermediate layer of the discriminator for unlabeled and generated inputs, and $N$ denote the batch size. The generator's loss is given by

$$l_{\mathcal{G}} = \frac{1}{N} \sum_{i=1}^{N} \left| \bar{x}_{\text{ul}}^{(i)} - \bar{x}_{\text{gen}}^{(i)} \right|. \tag{6}$$

The training procedure of the proposed framework is described in Algorithm 1. $D/C$'s parameters $\theta_{d/c}$ are updated w.r.t. its outputs on $x_l$, $x_{\text{ul}}$, and $x_{\text{gen}}$. This step is followed by an update of the generator parameters $\theta_g$ w.r.t. the distance between the intermediate activations for $x_l$ and $x_{\text{gen}}$ using the following equation:

$$\nabla\theta_g \frac{1}{m}\sum_{i=1}^{m}\left|\phi\left(I\left(x_{\text{ul}}^{(i)}\right)\right)-\phi\left(I\left(\mathcal{G}\left(z^{(i)}\right)\right)\right)\right| \tag{7}$$

where $\phi$ stands for variance/mean. The minimization of this distance results in an approximation of the probability distribution of generated samples $p_{\text{data}}(x_{\text{gen}})$ to the distribution of unlabeled samples $p_{\text{data}}(x_{\text{ul}})$. Combined together, the value function $\mathcal{V}$ of the adversarial training is given by

$$\begin{aligned}\min_{G}\max_{\mathcal{D}/\mathcal{C}}&\ \mathcal{V}(\mathcal{D}/\mathcal{C},\mathcal{G})\\ &=\sum_{c=1}^{c} y_l^{(i)}\log\left(\mathcal{C}\left(x_l^{(i)}\right)\right)\\ &\quad+\log\left(\mathcal{D}\left(x_{\text{ul}}^{(i)}\right)\right)+\log\left(1-\mathcal{D}\left(\mathcal{G}\left(z^{(i)}\right)\right)\right)\\ &\quad+\frac{1}{N}\sum_{i=1}^{N}\left|\bar{x}_{\text{ul}}^{(i)}-\bar{x}_{\text{gen}}^{(i)}\right|.\end{aligned} \tag{8}$$

**Algorithm 1** Training Algorithm for Semisupervised GAN. Networks Are Trained With a Learning Rate of 0.0002 and a Batch Size of $m = 128$. $\mathcal{C}$ = Classifier, $\mathcal{D}$ = Discriminator, $\mathcal{G}$ = Generator, and $I$ = Intermediate Output of the $\mathcal{D}/\mathcal{G}$ Network

**Input:** *n_iter*: number of total iterations

**for** *i*=1 **to** ***n_iter*** **do**

Sample $m$ noise samples $\{z^m\}_m^{i=1}$ from noise prior $\mathcal{P}_g(z)$.

Sample $m$ randomly labeled samples $\{(x_l{}^m, y_l{}^m)\}_m^{i=1}$ from the labeled set.

Sample $m$ random unlabeled samples $\{x_{ul}{}^m\}_m^{i=1}$ from the unlabeled set.

With:

$$\mathcal{D}(x)=\frac{\mathcal{Z}(x)}{\mathcal{Z}(x)+1},\ \text{where } \mathcal{Z}(x)=\sum_{k=1}^{K} exp[\mathcal{C}(x)]$$

Update the discriminator/classifier using *Adam* optimizer by:

$$\nabla\theta_d\frac{1}{m}\sum_{i=1}^{m}[\log((x_{ul}^{(i)}))+\log(1-\mathcal{D}(\mathcal{G}(z^{(i)})))+\sum_{c=1}^{c} y_l^{(i)}\log(\mathcal{C}(x_l^{(i)}))]$$

Update the generator using Adam optimizer by:

$$\nabla\theta_g\frac{1}{m}\sum_{i=1}^{m}|\phi(I(x_{ul}^{(i)}))-\phi(I(\mathcal{G}(z^{(i)})))|$$

**end for**

The framework is trained using a batch size of 128 and the *Adam* optimizer with a learning rate of $2e-4$. One-sided label smoothing with a factor of 0.1 is applied for unlabeled samples. The classifier's performance on the validation set is used as a stopping criterion. The experiments are conducted on an NVIDIA Tesla V100 GPU using the TensorFlow framework. The number of parameters for each dataset is shown in Table II. A code sample of the architecture is publicly available at https://github.com/Hazarzi/SSGAN_HAR.

## D. Complexity

This section presents the time complexity of the SSGAN architecture. Following [57], the time complexity of $d$ consecutive convolutional layers with $n$ flat kernels of size $s$ can be expressed as

$$O\left(\sum_{l=1}^{d} n_{l-1}s_l n_l m_l^2\right) \tag{9}$$

where $m_l$ is the spatial size of the output feature map. Note that $s_l$ becomes $s_l^2$ when both kernel dimensions are greater than one. Similarly, the time complexity of $l$ feed-forward layers with $j$ inputs, where the $i$th hidden layer contains $h_i$ hidden neurons, and $o$ output neurons is given by

$$O\left(jh_1+h_l o+\sum_{i=1}^{l-1}h_i h_{i+1}\right). \tag{10}$$

*1) Discriminator/Classifier Complexity:* Assuming that the pooling and activation operations have a negligible impact on the time complexity, the time complexity for the $D/C$ network with 4 convolutional layers ($C1$, $C2$, $C3$, and $C4$), three feed-forward layers ($D1$, $D2$, and $D3$), $t$ iterations, and $j$ epochs are given by

$$O_{D/C}\left(tj\left(\underbrace{\sum_{l=1}^{2}n_{l-1}s_l n_l m_l^2}_{C1,C2}+\underbrace{\sum_{l=2}^{4}n_{l-1}s_l 2n_l m_l^2}_{C3,C4}+\underbrace{2n_l m_l^2 h_1+h_l k\sum_{i=1}^{M-1}h_1 h_{i+1}}_{D1,D2,D3}\right)\right). \tag{11}$$

*2) Generator Complexity:* Following the equations presented earlier, with an input noise $z$, the time complexity of the generator that consists of a single feed-forward layer ($D1$) followed by four convolutional layers ($C1$, $C2$, $C3$, and $C4$) is given by

$$O_G\left(tj\left(\underbrace{zm_1+m_1k}_{D1}+\underbrace{n_0s_1 2n_1m_1^2}_{C1}+\underbrace{n_1s_2 4n_2m_2^2}_{C2}+\underbrace{n_2/2s_3 8n_3m_3^2}_{C3}+\underbrace{n_3/2s_4 16n_4m_4^2}_{C4}\right)\right). \tag{12}$$

The overall time complexity of the SSGAN architecture $O_{\text{SSGAN}}$ is, therefore, $O_{\text{SSGAN}} = O_{D/C} + O_G$. Additionally, as three forward passes of $D/C$ and one forward pass of $G$ are required during the training phase, the complexity of training SSGAN is approximately four times greater than the classification.

TABLE II
NUMBER OF PARAMETERS AND FLOPS W.R.T. THE INPUT DIMENSIONS

| Dataset | Data shape | Network | # Parameters | FLOPS |
|---|---|---|---|---|
| PAMAP2 | (90,40) | D/C<br>G | 14.580.172<br>3,453,444 | 0.622 G<br>0.561 G |
| Opportunity | (90,113) | D/C<br>G | 40,891,973<br>9,340,164 | 1.76 G<br>1.58 G |
| LISSI | (90,50) | D/C<br>G | 18,184,136<br>4,259,844 | 0.778 G<br>0.701 G |

Table II presents the number of parameters and FLOPS for the $D/C$ and $G$ networks. It can be seen that the number of parameters $m$ and FLOPS increases linearly w.r.t. the number of features of the input data.

## V. DATA PREPROCESSING

The datasets used in this study, as well as the preprocessing steps applied to them, are described in this section.

### A. Datasets

Experiments are carried out on three HAR datasets: publicly available opportunity/locomotion [58], PAMAP2 [53], and finally, a third dataset created by LISSI at the University of Paris-Est Créteil (UPEC), Créteil, France. Each dataset differs in terms of the number of participants, sensors, types of activities, and the number of activity classes, which allows for evaluating the effectiveness of the proposed approach under different conditions. In this work, only the recordings from wearables are used. It should be noted that all data in each dataset are annotated. Consequently, labeled and unlabeled subsets are prepared depending on different experimental conditions. For all three datasets, data are collected from healthy adult participants. The information related to activities and class distributions for each dataset is available in Fig. 2.

*1) PAMAP2 Dataset:* The PAMAP2 dataset is created within the context of the PAMAP2 project [53] and includes 18 different activities of daily living performed by nine participants. Data are recorded from three IMUs placed on the wrist, chest, and ankle, and a heart rate sensor, resulting in 52 features. The dataset is sampled at 100 Hz. The PAMAP2 dataset includes high-level and medium-level activities. In the protocol setting, the participants are instructed to carry out 12 activities (*lie, sit, stand, walk, run, cycle, Nordic walk, iron, vacuum cleaner, rope jump, ascend stairs, and descend stairs*) and six optional activities (*watch TV, computer work, drive car, fold laundry, clean house, and play soccer*). Only three participants performed all 12 activities from the protocol setting. In this work, following [22], 40 features and all 12 activities available in the protocol setting are used. The dataset is downsampled to 33.3 Hz by removing one-third of the data points. Data from participants 5 and 6 are set aside for validation and testing, respectively.

*2) Opportunity-Locomotion Dataset:* The Opportunity dataset is a large-scale multimodal dataset of human activities [58] simulating the daily life environment inside a studio flat and is annotated for both high/level and low/level activities. Recorded activities include modes of locomotion (*sitting, standing, walking, and lying*), actions of the right and left hand (e.g., *reach, grasp, and release*), interactions with different objects (e.g., *door and switch*), high/level activities

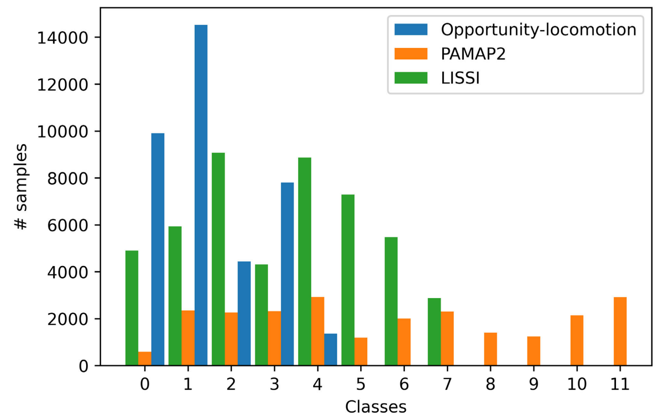


Fig. 2. Class distributions after data segmentation with overlap.

(e.g., *relaxing, coffee time, and sandwich time*), and finally, mid/level gestures generated from low/level hand actions (e.g., *open door*). The dataset is collected from four participants at a sampling frequency of 30 Hz by using wearable IMUs, objects equipped with sensors, and ambient sensors. The final dataset includes 242 features. The Opportunity-locomotion dataset includes four high/level activities (*standing, walking, sitting, and lying*) and a *null* class representing irrelevant activities. Following [5], 113 features and all five activities are selected. ADL3 sessions from participants 2 and 3 are used for validation, and ADL4 and ADL5 sessions from the same participants are used for testing.

*3) LISSI Dataset:* This dataset is created by the LISSI laboratory within the context of the European Medolution research project for the monitoring of rehabilitation self-exercises. Samples are collected using wearable IMUs at a sampling frequency of 60 Hz from 16 participants following a predefined protocol. The data recorded from wearables correspond to 110 features. The dataset has eight high-level (e.g., *standing, walking, and lying*) and 39 low-level activities (e.g., *holding the bent knee, stretching hands up, and retract*), including ten different categories of breaks. The dataset is recorded while reflecting real-life settings. In this study, eight high-level activities and 50 features that are common for every participant are used. The data from participants 4 and 15 are used for validation, and the data from participants 5 and 16 are used for testing. The rest of the participants' data are used for training purposes.

### B. Preprocessing

Missing data are forward and backward-filled using the last and first known values. Datasets are filtered using Butterworth high/pass and low/pass filters. For the PAMAP2 dataset, a high/pass filter with $Fc_{\text{high}} = 0.001$ is applied to all features except the temperature and heart rate, which are filtered using a low-pass filter with $Fc_{\text{low}} = 0.1$. For Opportunity/locomotion, features are only high/pass filtered using $Fc_{\text{high}} = 0.001$. Finally, the features of the LISSI dataset are filtered using a band/pass filter with $Fc_{\text{low}} = 0.01$ and $Fc_{\text{high}} = 1$. Samples of the PAMAP2 and Opportunity-locomotion datasets are scaled between $-1$ and 1 using maximum and minimum sensor values. The samples of the LISSI dataset are scaled between

TABLE III
SUMMARY OF THE EXPERIMENTAL SETUP

| Dataset | # subjects | # labels | # features | Validation | Test | Window size/Overlap |
|---|---|---|---|---|---|---|
| PAMAP2 | 9 | 12 | 40 | Subject 5 | Subject 6 | 1.5 s/67% |
| Opportunity | 4 | 5 | 113 | Subject 2 ADL3, Subject 3 ADL3 | Subject 2 ADL4, 5, Subject 3 ADL4, 5 | 3 s/67% |
| LISSI | 16 | 8 | 50 | Subject 5, Subject 17 | Subject 6, Subject 20 | 1.5 s/67% |

−1 and 1 using the statistics of the training data. Datasets are segmented by applying sliding windows with overlap, and transitions between activities are removed. The window lengths are 3, 3, and 1.5 s for the Opportunity-locomotion, PAMAP2, and LISSI datasets, respectively, corresponding to 90 timesteps for each dataset.

## VI. RESULTS

This section presents the results obtained from the evaluation of the proposed semisupervised GAN framework under different experimental conditions. Additionally, it provides a detailed description of the experimental conditions used in the evaluation of the proposed approach.

The effectiveness of the proposed framework is tested under four conditions reflecting real-world SSL scenarios: 1) using labeled data from only one subject; 2) reducing the amount of labeled data; 3) reducing the number of classes; and 4) combining different feature subsets based on sensor locations. Table III summarizes the experimental setup. In this study, the weighted F1-score is used as an evaluation metric to take into account the class imbalance, which is defined by

$$F_w = 2\sum_c \frac{N_c}{N_{\text{total}}}\frac{\text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \quad (13)$$

### A. Analysis of the Framework During Training

A semisupervised training experiment is conducted, as explained in Section VI-B1, on the PAMAP2 dataset, using the labeled data from the first subject and unlabeled data from the rest of the training subjects. Fig. 3 shows the evolution of loss values and the validation performance for the discriminator/classifier and generator networks during the training. It can be seen that the loss values stabilize at approximately 25 epochs, corresponding to convergence as validation performance stops increasing. Note that the loss values of the generator and discriminator/classifier networks are not on the same scale as different loss functions are used (see Section IV-C). Similar training dynamics are consistently observed during training under different experimental conditions. A sample generated during the experiment, as shown in Fig. 4, demonstrates that the generator can produce signals for different sensors; however, unlike images, the visual assessment of signal quality is challenging. In addition, the feature matching technique significantly reduces the quality of the generated samples [9]. Indeed, this reduction is expected, as feature matching forces the generator to match the probability distribution of the real data instead of matching the data itself. As described in Section IV-B, this prevents the mode collapse by regularizing the generator and improves the classification performance.

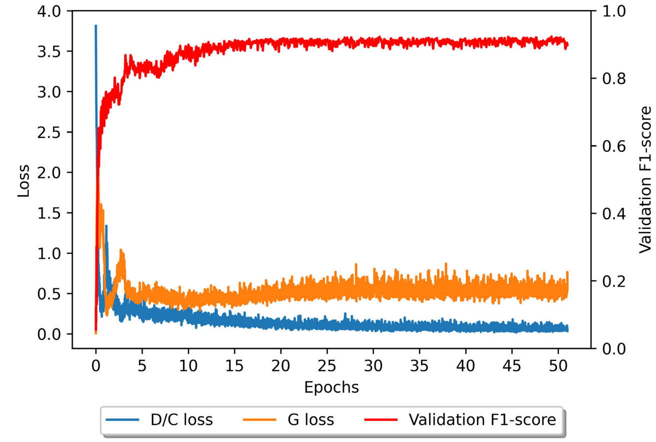


Fig. 3. Loss graph of the discriminator/classifier (D/C) and generator networks (G) from the training of the semisupervised GAN on the PAMAP2 dataset.

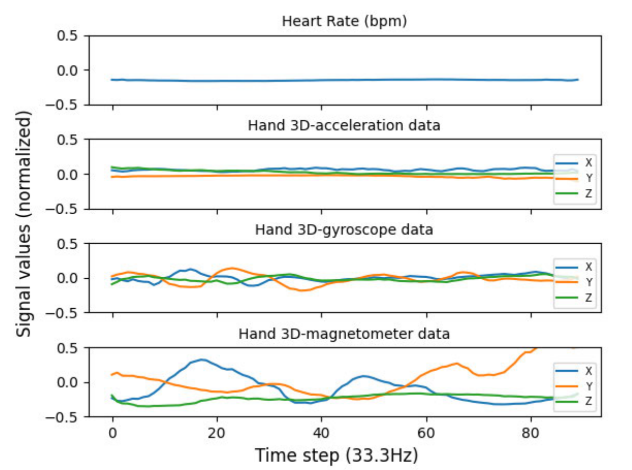


Fig. 4. Sample created by the generator network. Only ten features from the PAMAP2 dataset are visualized.

No mode collapse is observed during the visual evaluation of the generated samples.

### B. Evaluation of the Proposed Framework

The differences in classification performance associated with SSL are demonstrated by evaluating the proposed framework under different experimental conditions and comparing its performance to that of a supervised CNN. In this regard, the *supervised* column in Sections VI-B1–VI-B4 refers to the results of supervised CNNs trained on the same labeled set as the semisupervised GANs. The CNN models use the same architecture as the discriminator/classifier network. The validation and test sets used for evaluation are available in Table III.

TABLE IV
SUPERVISED CNN AND BI-LSTM CLASSIFICATION PERFORMANCE ON ALL LABELED TRAINING DATA IN TERMS OF F1-SCORE

| | Previous Results | | Our Results | |
|---|---|---|---|---|
| Dataset | CNN[46] | bi-LSTM[22] | CNN | bi-LSTM |
| PAMAP2 | 0.909 | 0.868 | 0.915 | 0.872 |
| Opportunity | 0.896 | - | 0.882 | 0.878 |
| LISSI | - | - | 0.910 | 0.899 |

In this section, the performance of the CNN architecture used within the proposed SSGAN framework is validated by comparing its performance to a 256-unit bi-LSTM, as well as to the CNN and LSTM architectures proposed in [22] and [46]. The same training, validation, and testing sets are used in [46] and [22], respectively. The aim of this step is to verify that the CNN model performs well in a supervised setting and is capable of achieving results similar to previous work on supervised HAR with CNNs and RNNs. In contrast, no comparable studies could be identified for the bi-LSTM using the Opportunity-locomotion dataset. The performance of the classifiers on the LISSI dataset is not compared with previous studies as it is not publicly available. The results of the training are shown in Table IV.

After training the CNN model used within the discriminator/classifier on the PAMAP2 dataset, the resulting classification performance is 0.6% higher than the CNN in [46] in terms of the weighted F1-score. For the Opportunity-locomotion dataset, this performance is 0.14% lower. The classification performance of the bi-LSTM network on the PAMAP2 dataset is 0.4% higher than the bi-LSTM in [22] in terms of the average F1-score. The CNN classifier outperforms the bi-LSTM networks on each dataset. These results confirm that the supervised models used in this study achieve similar performance to that of previous studies. The slight differences in classification performance compared with previous studies may be attributed to the additional step of signal filtering. These results also represent the upper limit of achievable performance through supervised training for the rest of the experiments in this study.

In Sections VI-B.1–VI-B.4, the effectiveness of the proposed framework is tested under four conditions reflecting real-world SSL scenarios.

*1) Impact of Cross-Subject Training:* For the first experimental condition, the semisupervised GANs are trained in a cross-subject context using labeled data from a single participant. The training data from the other participants are used as unlabeled data. The objective is to investigate the impact of using labeled and unlabeled data from different participants on the performance of the proposed semisupervised GAN framework. The results are reported in Table V and Fig. 5. The classification accuracy scores of the proposed framework for different types of activities are reported in Table VI.

For each dataset, the semisupervised GANs consistently achieved better classification performance than the supervised models trained on the same labeled data. Semisupervised training resulted in average classification performance of 87.7%, 86.5%, and 88.4%, corresponding to average improvements of 10.9%, 8.3%, and 12.1% in terms of the weighted F1-score for the PAMAP2, Opportunity-locomotion, and LISSI datasets, respectively.

TABLE V
SUPERVISED CNN AND SEMISUPERVISED GAN CLASSIFICATION PERFORMANCE IN THE CROSS-SUBJECT SETTING. THE BEST RESULTS ARE MARKED IN BOLD

| Dataset | Subject | Supervised | SSGAN | Δ (%) |
|---|---|---|---|---|
| PAMAP2 | S1 | 0.762 | **0.912** | 14.9 |
| | S8 | 0.772 | 0.843 | 7 |
| | Mean | 0.767 | 0.877 | 10.9 |
| Opportunity | S1 | 0.824 | **0.902** | 7.7 |
| | S2 | 0.811 | 0.889 | 7.7 |
| | S3 | 0.688 | 0.794 | 10.5 |
| | S4 | 0.802 | 0.875 | 7.2 |
| | Mean | 0.782 | 0.865 | 8.3 |
| LISSI | S1 | 0.716 | 0.913 | 19.7 |
| | S2 | 0.661 | 0.862 | 20.1 |
| | S3 | 0.762 | **0.924** | 16.2 |
| | S6 | 0.783 | 0.888 | 10.5 |
| | S7 | 0.729 | 0.886 | 15.7 |
| | S8 | 0.664 | 0.820 | 15.6 |
| | S9 | 0.790 | 0.905 | 11.5 |
| | S10 | 0.822 | 0.911 | 8.9 |
| | S11 | 0.782 | 0.787 | 0.5 |
| | S12 | 0.866 | 0.919 | 5.3 |
| | S13 | 0.782 | 0.894 | 11.2 |
| | S14 | 0.805 | 0.903 | 9.8 |
| | Mean | 0.763 | 0.884 | 12.1 |

The results also reveal significant differences in classification performance between subjects. In some cases, such as the semisupervised training with labeled data of Subject 3 from the LISSI dataset, the proposed framework outperforms the supervised CNN trained with all available labeled data (see Table IV). For the PAMAP2 dataset, training the semisupervised GANs with the labeled data of Subject 1 yields the highest classification performance with an F1-score of 91.2%, which is 6.9% higher than that of Subject 8. For the Opportunity dataset, the labeled data of Subject 1 yields the best performance with an F1-score of 90.2%, which is 10.8% higher than the worst-performing subject. Finally, Subject 3 yields the best performance for the LISSI dataset with a weighted F1-score of 92.4%, which is 13.7% higher than the worst-performing subject. These differences are related to intersubject variability within datasets.

The study of classification performance for different types of activities in the PAMAP2 dataset reveals that most of the misclassified examples belong to *ascending stairs* (see Table VI). This phenomenon is also reported in a previous study [31] and is related to the similarity between the *ascending stairs* and *descending stairs* activities. For the Opportunity-locomotion dataset, the majority of incorrect classifications belong to the *walking* activity. Finally, for the LISSI dataset, most of the misclassified instances belong to the *sitting* activity. The impact of excluding the different activities is studied in Section VI-B3.

*2) Impact of the Amount of Labeled Data:* The performance of the proposed framework is evaluated by reducing the amount of labeled data to 50, 100, 200, 500, 1000, 2000, and 5000 instances that are randomly selected in a stratified manner to represent true class distributions. The training data from all participants are pooled, and after the labeled data are selected, the remaining training data are used as the unlabeled set. In addition, the semisupervised training results are also compared with the supervised learning performance achieved using all labeled data presented in Section VI-A. This

TABLE VI
SEMISUPERVISED GAN CLASSIFICATION ACCURACY SCORES FOR DIFFERENT TYPES OF ACTIVITIES

| PAMAP2 | | Opportunity | | LISSI | |
|---|---|---|---|---|---|
| Activity | Acc. | Activity | Acc. | Activity | Acc. |
| A. Stairs | 0.61 | Null | 0.85 | Warm Up | 0.92 |
| Standing | 0.83 | Walking | 0.69 | Sitting | 0.71 |
| Sitting | 0.86 | Standing | 0.96 | Kneeling | 0.90 |
| D. Stairs | 0.87 | Sitting | 0.99 | Walking | 0.95 |
| Ironing | 0.93 | Lying | 0.99 | Lying | 0.96 |
| Vacuum Cleaning | 0.94 | | | Sit. to Std. | 0.97 |
| Running | 0.94 | | | Standing | 0.97 |
| Cycling | 0.97 | | | Relaxation | 0.97 |
| Nordic Walking | 0.98 | | | | |
| Lying | 0.98 | | | | |
| Walking | 0.98 | | | | |

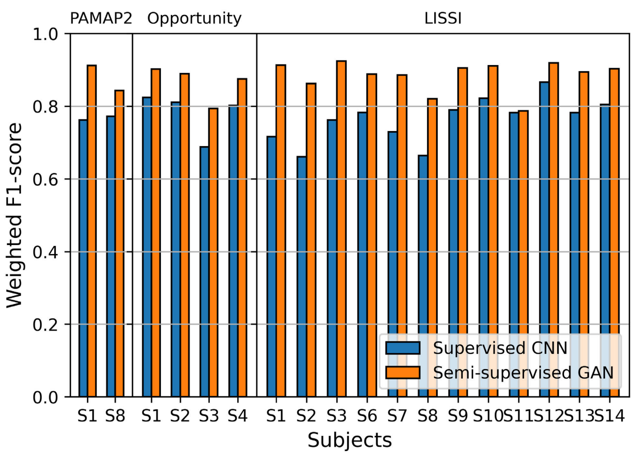


Fig. 5. Comparison of semisupervised GAN and supervised CNN classification performance in a cross-subject setting.

is done to determine the minimum amount of labeled data needed to achieve semisupervised performance comparable to that of fully supervised training. The results are presented in Table VII and Fig. 6.

As in the previous experiment, semisupervised GANs yield higher classification performance than the supervised models for each dataset. For the PAMAP2 dataset, using 500 labeled samples yields a weighted F1-score of 91%, which is 4.4% better than supervised training and 0.5% worse than training with all the annotated data. Using 5000 labeled samples from Opportunity/locomotion yields a weighted F1-score of 91%, which is 5.6% better than the supervised model and also 2.8% better than the supervised model trained with all labeled data. With only 1000 labeled samples, the semisupervised GAN is able to outperform the supervised training with all annotated data by 1.2%. Finally, for the LISSI dataset, using 5000 labeled instances results in a weighted F1-score of 92.9%, an improvement of 5.6% over the supervised model, and an outperformance of 1.9% over training with all annotated data.

The results show that the amount of annotated data required to achieve the classification performance of supervised training on all annotated data varies by dataset. Using only 500 annotated data from the PAMAP2 and Opportunity-locomotion datasets and 1000 annotated data from the LISSI dataset, the semisupervised GANs achieve classification performance comparable to supervised classifiers using all available training data. The results also indicate an inverse

TABLE VII
SUPERVISED CNN AND SEMISUPERVISED GAN CLASSIFICATION PERFORMANCE WITH DIFFERENT AMOUNTS OF LABELED DATA. THE BEST RESULTS ARE MARKED IN BOLD. [1]RESULTS ON ALL LABELED DATA FROM SECTION VI-B

| Dataset | # labeled | Supervised | SSGAN | Δ (%) |
|---|---|---|---|---|
| PAMAP2 | 50 | 0.566 | 0.731 | 16.5 |
| | 100 | 0.633 | 0.794 | 16.1 |
| | 200 | 0.733 | 0.889 | 15.6 |
| | 500 | 0.866 | **0.910** | 4.4 |
| | 1000 | 0.901 | 0.908 | 0.7 |
| | 2000 | 0.888 | 0.908 | 2 |
| | 5000 | 0.894 | **0.910** | 1.6 |
| | 17273[1] | 0.915 | - | - |
| Opportunity | 50 | 0.734 | 0.795 | 6.2 |
| | 100 | 0.723 | 0.774 | 5.1 |
| | 200 | 0.782 | 0.832 | 5 |
| | 500 | 0.821 | 0.881 | 6 |
| | 1000 | 0.834 | 0.894 | 6 |
| | 2000 | 0.847 | 0.884 | 3.7 |
| | 5000 | 0.853 | **0.910** | 5.7 |
| | 16641[1] | 0.882 | - | - |
| LISSI | 50 | 0.501 | 0.680 | 17.9 |
| | 100 | 0.642 | 0.807 | 16.5 |
| | 200 | 0.692 | 0.856 | 16.4 |
| | 500 | 0.756 | 0.891 | 13.5 |
| | 1000 | 0.824 | 0.917 | 9.3 |
| | 2000 | 0.833 | 0.897 | 6.4 |
| | 5000 | 0.873 | **0.929** | 5.6 |
| | 34666[1] | 0.910 | - | - |

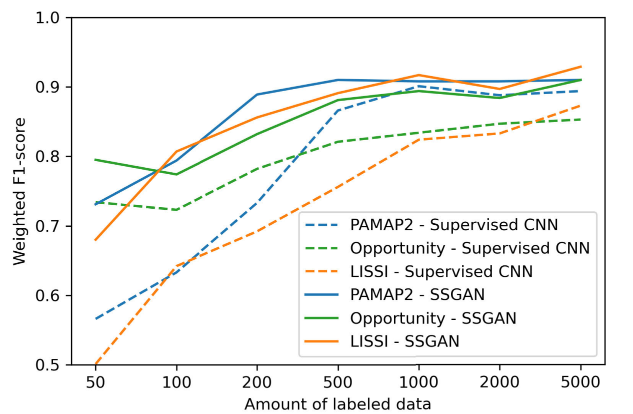


Fig. 6. Effects of different amounts of labeled data on the classification performance of semisupervised GANs and supervised CNNs.

correlation between the performance of the supervised classifiers and the percentage improvement achieved by the semisupervised training. This relationship may indicate an upper bound on achievable classification performance, resulting in diminishing returns as the number of labeled data increases.

*3) Impact of Number of Classes on Model Performance:* In the third experimental condition, the impact of the number of classes on classification performance is studied in a cross/subject context as in the first configuration. However, labeled data are selected only from the participants with the best classification performance in the first experimental condition, corresponding to the first participants from the PAMAP2 and Opportunity-locomotion datasets and the third participant from the LISSI dataset. The rest of the participants' data are used as the unlabeled set. The activities are

TABLE VIII
SUPERVISED CNN AND SEMISUPERVISED GAN CLASSIFICATION PERFORMANCE ON DIFFERENT NUMBERS OF LABELS. THE BEST RESULTS ARE MARKED IN BOLD. [1]RESULTS WITH ALL CLASSES

| Dataset | # labels | Supervised | SSGAN | Δ (%) |
|---|---|---|---|---|
| PAMAP2 | 2 | 0.999 | **0.996** | -0.3 |
| | 3 | 0.978 | 0.990 | 1.2 |
| | 4 | 0.940 | 0.987 | 4.7 |
| | 5 | 0.934 | 0.973 | 3.9 |
| | 6 | 0.921 | 0.975 | 5.4 |
| | 7 | 0.925 | 0.961 | 3.6 |
| | 8 | 0.870 | 0.950 | 8.0 |
| | 9 | 0.880 | 0.962 | 8.2 |
| | 10 | 0.841 | 0.891 | 5.0 |
| | 11 | 0.843 | 0.904 | 6.1 |
| | 12[1] | 0.762 | 0.912 | 14.9 |
| Opportunity | 2 | 0.960 | **0.999** | 3.8 |
| | 3 | 0.943 | 0.993 | 5.0 |
| | 4 | 0.938 | 0.987 | 4.9 |
| | 5[1] | 0.824 | 0.902 | 7.7 |
| LISSI | 2 | 0.957 | **0.997** | 4.0 |
| | 3 | 0.929 | **0.997** | 6.8 |
| | 4 | 0.866 | 0.984 | 11.8 |
| | 5 | 0.834 | 0.979 | 14.5 |
| | 6 | 0.821 | 0.958 | 13.7 |
| | 7 | 0.739 | 0.936 | 19.7 |
| | 8[1] | 0.762 | 0.924 | 16.2 |

TABLE IX
SUPERVISED CNN AND SEMISUPERVISED GAN CLASSIFICATION PERFORMANCE FOR DIFFERENT IMU LOCATIONS. THE BEST RESULTS ARE MARKED IN BOLD. [1]SUPERVISED CLASSIFICATION PERFORMANCES FROM SECTION VI-B1. [2]SINGULAR ACCELEROMETERS ARE EXCLUDED

| Dataset | # Location | Supervised | SSGAN | Δ (%) |
|---|---|---|---|---|
| PAMAP2 | Wrist | 0.528 | 0.778 | 25 |
| | Chest | 0.544 | 0.771 | 22.7 |
| | Ankle | 0.573 | 0.786 | 21.3 |
| | Wrist-Chest | 0.740 | 0.901 | 16.1 |
| | Wrist-Ankle | 0.689 | 0.842 | 15.3 |
| | Chest-Ankle | 0.702 | 0.800 | 9.8 |
| | All locations[1] | 0.762 | **0.912** | 14.9 |
| Opportunity | Arms | 0.697 | 0.756 | 6 |
| | Back | 0.791 | 0.854 | 6.2 |
| | Shoes | 0.648 | 0.713 | 6.5 |
| | Arms-Back | 0.764 | 0.847 | 8.2 |
| | Arms shoes | 0.792 | 0.878 | 8.6 |
| | Back shoes | 0.752 | 0.805 | 5.2 |
| | All locations[2] | 0.766 | **0.884** | 11.7 |
| LISSI | Wrists | 0.663 | 0.852 | 18.9 |
| | Chest | 0.554 | 0.777 | 22.3 |
| | Ankles | 0.670 | 0.885 | 21.5 |
| | Wrists-Chest | 0.754 | **0.929** | 17.5 |
| | Wrists-Ankles | 0.747 | 0.895 | 14.9 |
| | Chest-Ankles | 0.714 | 0.904 | 19 |
| | All locations[1] | 0.762 | 0.924 | 16.2 |

removed one by one, based on the performance of the semisupervised GAN on the different activities shown in Table VI. Exceptionally, for the Opportunity-locomotion and LISSI datasets, the activities *null* and *warm-up* are removed first, respectively, as they represent activities that are irrelevant to the activities of interest. Similarly, for the PAMAP2 dataset, the activity *jump rope* is removed first as it is not available in the test set. The results are available in Table VIII and Fig. 7.

In almost all cases, with the exception of the PAMAP2 dataset with two labels, the semisupervised GANs achieved higher classification performance than the supervised models. The most significant performance improvement over using all labels is achieved after removing the *null* activity from the Opportunity-locomotion dataset, resulting in a weighted F1-score of 98.7% after semisupervised training, an improvement of 4.9%. By removing three (*nordic walking, cycling, and running*) and two (*warm-up* and sitting to standing) activities from the PAMAP2 and LISSI datasets, respectively, the proposed framework achieved weighted F1-scores above 95% for both datasets. In these cases, the improvement over the CNN classifiers is 8.2% for the PAMAP2 dataset and 13.7% for the LISSI dataset.

The results indicate that some activities are more difficult to classify. As expected, removing activities based on the number of misclassifications improved the performance of the supervised model and the semisupervised GAN. Meanwhile, removing the *rope jumping* and *ascending stairs* activities from the PAMAP2 dataset reduced the performance of the proposed framework by 2.1%. Although the *rope jumping* activity is removed first as it is not present in the test set, it may be important for the correct classification of other activities. The reason for the decrease in performance after removing the *ascending stairs* activity needs further investigation. Finally, in the Opportunity-locomotion dataset, the negative impact of *null* activity on model performance is also reported in a comparative study [58].

Similar to the results in Section VI-B2, the relative increase in classification performance from semisupervised training is lower when supervised performance is already high. Considering the PAMAP2 dataset, training the proposed framework with 12 activities resulted in an improvement of 14.9%, while for three activities, the improvement is only 1.2%, and similar results are observed with the other datasets.

*4) Impact of IMU Positions on Model Performance:* For the last experimental condition, the framework is evaluated under different feature combinations by selecting single or multilocation feature sets based on the IMU positions. For the Opportunity-locomotion dataset, triaxial accelerometer readings from singular accelerometers are excluded, as the IMUs already have built-in accelerometers. For the PAMAP2 dataset, the heart rate feature is included in each subset, and data from IMUs located on the wrist, chest, and ankle are used. For the Opportunity-locomotion dataset, IMU data from the right and left lower arms, back, and right and left shoes are used. Finally, IMU data from the right and left wrists, chest, and right and left ankles are used for the LISSI dataset. The features from the IMUs located on both the left and right sides of the participants are kept for the Opportunity-locomotion and LISSI datasets, as the laterality of participants is unknown. The evaluation parameters are the same as in the previous experiment. The performance of the supervised models and the semisupervised GAN for different groups of features are presented in Table IX and Fig. 8.

For the PAMAP2 dataset, in a multilocation configuration, the three IMU combination performs best. The wrist–chest combination yields a weighted F1-score of 90.1%, which is 1.1% lower than the three-location combination. In the single-location case, all locations provide similar classification performance, resulting in a weighted F1-score of approximately 77% after semisupervised training. Similarly, for the Opportunity dataset, the combination of all locations performs

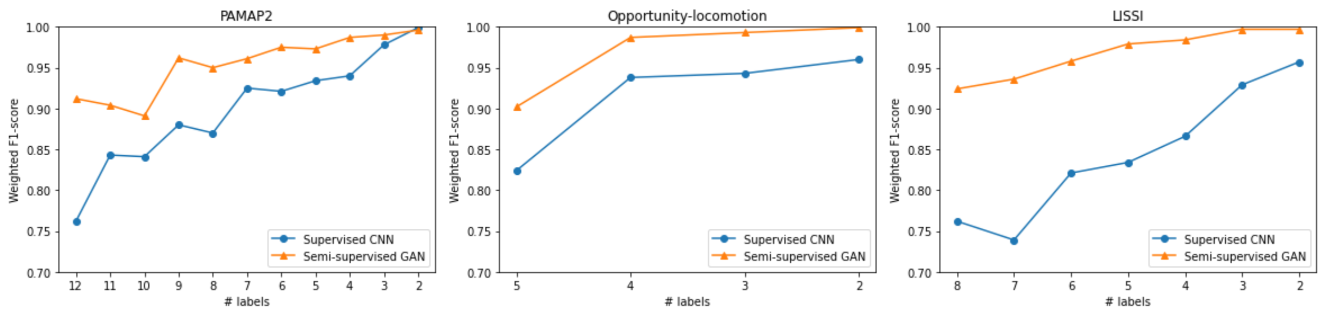


Fig. 7. Effects of different numbers of labels on the classification performance of semisupervised GANs and supervised CNNs.

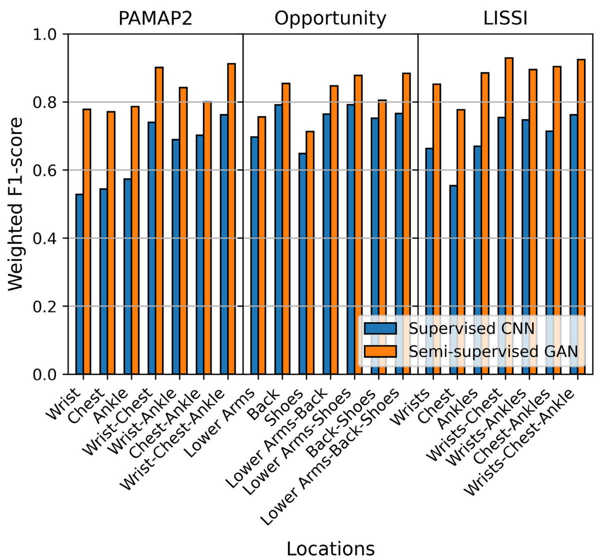


Fig. 8. Comparison of semisupervised and supervised classification performance for different IMU locations.

best, while the arm–shoe combination yields a weighted F1 score of 87.8%, which is 0.6% lower than all locations combined. For a single location, the back IMU also performed comparably, corresponding to 85.4% in terms of weighted F1 score. For the LISSI dataset, the wrist–chest combination outperformed the all-location combination by 0.5%, resulting in an F1-score of 92.9%. For a single location, the ankle IMUs outperformed, with an F1-score of 85.2%.

The results show that the combination of two or three locations provided the best classification for each dataset, which is also demonstrated in [59]. In some cases, depending on the location of the IMUs, using two locations is sufficient to achieve similar performance to using all features, such as the wrist–chest combination of PAMAP2 and LISSI datasets that have medium to high activity levels. It is also shown in [60] that the wrist–chest combination in the PAMAP2 dataset provided good classification performance and that the ankle IMU had less impact on classification performance. The single-location configuration provided inferior classification performance in almost all cases. Overall, these results indicate that good classification performance can be achieved using fewer features. It should be noted that the presence of the *null* activity in the Opportunity-locomotion dataset can negatively impact classification performance for different sensor locations. A limitation of this experiment is that the combination of left and right IMUs can potentially induce bias in the final results. A previous study also showed that using sensors on the nondominant hand for low-level activities, such as *dining*, can have a negative impact on classification performance [61]. In addition, it can be argued that laterality has less impact on final performance if only high-level activities are present.

TABLE X
RESULTS FROM THE PREVIOUS STUDIES ON SSL USING THE PAMAP2 AND OPPORTUNITY-LOCOMOTION DATASETS. THE SSL METHODS USED IN THESE STUDIES INCLUDE ADVERSARIAL VAEs [52], LADDER CNNs [62], ACTIVE LEARNING WITH kNN AND RANDOM FORESTS [63], SEMISUPERVISED CROSS-SUBJECT DOMAIN ADAPTATION WITH GANs [4] AND COLLABORATIVE SELF-SUPERVISED LEARNING [64]

| Dataset | Study | Amount of labeled data | Results (F1) |
|---|---|---|---|
| PAMAP2 | [52] | %20 | 78.63% (Acc) |
| | [62] | ∼ 0.29% | 54.9% |
| | [63] | 8% | 84.6% |
| | [4] | ∼ 11% | 83% |
| | [64] | 50% | 65.1% |
| Opportunity | [63] | 20% | 81% |
| | [64] | 10% | 55.6% |
| PAMAP2 | Current work | ∼ 2.9% (500 samples) | **91%** |
| Opportunity | Current work | ∼ 3% (500 samples) | **88.1%** |

## C. Comparison With Results of Previous Studies

Table X presents the results of previous studies on SSL with Opportunity-locomotion and PAMAP2 datasets. Compared with these studies, the proposed framework achieved higher classification performance. However, it should be highlighted that these results are influenced by heterogeneity between HAR studies due to differences in preprocessing steps, model architectures, and parameters, making direct comparisons difficult. Based on these previous results, it can also be argued that the most significant increase in performance can be attributed to the choice of initial models and preprocessing steps. Depending on the context, these choices may have more impact on the final performance than the proposed SSL algorithms. For instance, in [4], supervised training on labeled data from a single subject in the PAMAP2 dataset resulted in a weighted F1-score of 4%. In contrast, in this work, the lowest performance for a supervised model trained on a single subject from the PAMAP2 dataset is 76.2% in terms of the weighted F1-score. This demonstrates the potential impact of architectures and preprocessing steps.

## VII. Conclusion

This article proposes an alternative semisupervised GAN framework with temporal convolutions for HAR applications. By implementing deep convolutional neural networks with the proposed modifications, the framework is able to address several issues in conventional approaches in the HAR context, such as high dimensionality, scarcity of annotated data, and scalability. The evaluation of different datasets has revealed the robustness of the framework while reducing the bias in the reported results.

The effectiveness of the proposed framework is validated under four conditions reflecting real-world SSL scenarios while investigating the impact of cross-subject training, the amount of labeled data, the number of classes, and IMU positions on model performance. The results show the proposed framework's high classification performance and generalization ability compared with baseline methods. The performance of the semisupervised classifiers is also comparable to or better than that of supervised classifiers when a small number of annotated data are available.

One limitation of this study is the signal filtering that influenced the learning stability of the proposed framework, which can be addressed in future work. Other future directions could include the study of the generated signal probabilities to estimate and maximize the information content, and the implementation of attention mechanisms within the framework.

## References


[1] E. Ramanujam, T. Perumal, and S. Padmavathi, "Human activity recognition with smartphone and wearable sensors using deep learning techniques: A review," *IEEE Sensors J.*, vol. 21, no. 12, pp. 13029–13040, Mar. 2021.

[2] O. D. Lara and M. A. Labrador, "A survey on human activity recognition using wearable sensors," *IEEE Commun. Surveys Tuts.*, vol. 15, no. 3, pp. 1192–1209, 3rd Quart., 2013.

[3] J. Wang, T. Zhu, J. Gan, L. L. Chen, H. Ning, and Y. Wan, "Sensor data augmentation by resampling in contrastive learning for human activity recognition," *IEEE Sensors J.*, vol. 22, no. 23, pp. 22994–23008, Dec. 2022.

[4] E. Soleimani, G. Khodabandelou, A. Chibani, and Y. Amirat, "Generic semi-supervised adversarial subject translation for sensor-based activity recognition," *Neurocomputing*, vol. 500, pp. 649–661, Aug. 2022.

[5] F. Ordóñez and D. Roggen, "Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition," *Sensors*, vol. 16, no. 1, p. 115, Jan. 2016.

[6] M. Koosha, G. Khodabandelou, and M. M. Ebadzadeh, "A hierarchical estimation of multi-modal distribution programming for regression problems," *Knowl.-Based Syst.*, vol. 260, Jan. 2023, Art. no. 110129.

[7] J. E. Van Engelen and H. H. Hoos, "A survey on semi-supervised learning," *Mach. Learn.*, vol. 109, no. 2, pp. 373–440, 2020.

[8] I. Goodfellow et al., "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 27, 2014, pp. 1–9.

[9] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen, "Improved techniques for training GANs," in *Proc. 30th Int. Conf. Neural Inf. Process. Syst. (NIPS)*, New York, NY, USA, 2016, pp. 2234–2242.

[10] A. Odena, "Semi-supervised learning with generative adversarial networks," 2016, *arXiv:1606.01583*.

[11] C. Li, K. Xu, J. Zhu, and B. Zhang, "Triple generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1–11.

[12] Z. Dai, Z. Yang, F. Yang, W. W. Cohen, and R. Salakhutdinov, "Good semi-supervised learning that requires a bad GAN," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst. (NIPS)*, New York, NY, USA, 2017, pp. 6513–6523.

[13] L. Mescheder, A. Geiger, and S. Nowozin, "Which training methods for GANs do actually converge?" in *Proc. 35th Int. Conf. Mach. Learn.*, 2018, pp. 3481–3490.

[14] N. Martinez-Martin et al., "Ethical issues in using ambient intelligence in health-care settings," *Lancet Digit. Health*, vol. 3, pp. e115–e123, Feb. 2021.

[15] A. Jobin, M. Ienca, and E. Vayena, "The global landscape of AI ethics guidelines," *Nature Mach. Intell.*, vol. 1, no. 9, pp. 389–399, 2019.

[16] A. Borji, "Pros and cons of GAN evaluation measures," *Comput. Vis. Image Understand.*, vol. 179, pp. 41–65, Feb. 2019.

[17] M. H. Arshad, M. Bilal, and A. Gani, "Human activity recognition: Review, taxonomy and open challenges," *Sensors*, vol. 22, no. 17, p. 6463, Aug. 2022.

[18] R. Huan et al., "Human complex activity recognition with sensor data using multiple features," *IEEE Sensors J.*, vol. 22, no. 1, pp. 757–775, Jan. 2022.

[19] S. Zhang et al., "Deep learning in human activity recognition with wearable sensors: A review on advances," *Sensors*, vol. 22, no. 4, p. 1476, Feb. 2022.

[20] N. Gupta, S. K. Gupta, R. K. Pathak, V. Jain, P. Rashidi, and J. S. Suri, "Human activity recognition in artificial intelligence framework: A narrative review," *Artif. Intell. Rev.*, vol. 55, no. 6, pp. 4755–4808, Aug. 2022.

[21] L. Tong, H. Ma, Q. Lin, J. He, and L. Peng, "A novel deep learning bi-GRU-I model for real-time human activity recognition using inertial sensors," *IEEE Sensors J.*, vol. 22, no. 6, pp. 6164–6174, Mar. 2022.

[22] N. Y. Hammerla, S. Halloran, and T. Plötz, "Deep, convolutional, and recurrent models for human activity recognition using wearables," in *Proc. 25th Int. Joint Conf. Artif. Intell.* Palo Alto, CA, USA: AAAI Press, 2016, pp. 1533–1540.

[23] K. Xia, J. Huang, and H. Wang, "LSTM-CNN architecture for human activity recognition," *IEEE Access*, vol. 8, pp. 56855–56866, 2020.

[24] N. Dua, S. N. Singh, and V. B. Semwal, "Multi-input CNN-GRU based human activity recognition using wearable sensors," in *Computing*, vol. 103, no. 7, pp. 1461–1478, Jul. 2021.

[25] O. Nafea, W. Abdul, G. Muhammad, and M. Alsulaiman, "Sensor-based human activity recognition with spatio-temporal deep learning," *Sensors*, vol. 21, no. 6, p. 2141, Mar. 2021.

[26] Y. J. Luwe, C. P. Lee, and K. M. Lim, "Wearable sensor-based human activity recognition with hybrid deep learning model," *Informatics*, vol. 9, no. 3, p. 56, Jul. 2022.

[27] Y. Li and L. Wang, "Human activity recognition based on residual network and BiLSTM," *Sensors*, vol. 22, no. 2, p. 635, Jan. 2022.

[28] D. Bhattacharya, D. Sharma, W. Kim, M. F. Ijaz, and P. K. Singh, "Ensem-HAR: An ensemble deep learning model for smartphone sensor-based human activity recognition for measurement of elderly health monitoring," *Biosensors*, vol. 12, no. 6, p. 393, Jun. 2022.

[29] A. A. Varamin, E. Abbasnejad, Q. Shi, D. C. Ranasinghe, and H. Rezatofighi, "Deep auto-set: A deep auto-encoder-set network for activity recognition using wearables," in *Proc. 15th EAI Int. Conf. Mobile Ubiquitous Syst., Comput., Netw. Services*, Nov. 2018, pp. 246–253.

[30] Q. Ni et al., "Leveraging wearable sensors for human daily activity recognition with stacked denoising autoencoders," *Sensors*, vol. 20, no. 18, p. 5114, Sep. 2020.

[31] K. D. Garcia et al., "An ensemble of autonomous auto-encoders for human activity recognition," *Neurocomputing*, vol. 439, pp. 271–280, Jun. 2021.

[32] D. Thakur, S. Biswas, E. S. L. Ho, and S. Chattopadhyay, "ConvAE-LSTM: Convolutional autoencoder long short-term memory network for smartphone-based human activity recognition," *IEEE Access*, vol. 10, pp. 4137–4156, 2022.

[33] G. Khodabandelou, H. Moon, Y. Amirat, and S. Mohammed, "A fuzzy convolutional attention-based GRU network for human activity recognition," *Eng. Appl. Artif. Intell.*, vol. 118, Feb. 2023, Art. no. 105702.

[34] G. Khodabandelou, P.-G. Jung, Y. Amirat, and S. Mohammed, "Attention-based gated recurrent unit for gesture recognition," *IEEE Trans. Autom. Sci. Eng.*, vol. 18, no. 2, pp. 495–507, Apr. 2021.

[35] B. Khaertdinov, E. Ghaleb, and S. Asteriadis, "Deep triplet networks with attention for sensor-based human activity recognition," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun. (PerCom)*, Mar. 2021, pp. 1–10.

[36] Z. N. Khan and J. Ahmad, “Attention induced multi-head convolutional neural network for human activity recognition,” *Appl. Soft Comput.*, vol. 110, Oct. 2021, Art. no. 107671.

[37] Y. Tang, L. Zhang, Q. Teng, F. Min, and A. Song, “Triple cross-domain attention on human activity recognition using wearable sensors,” *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 6, no. 5, pp. 1167–1176, Oct. 2022.

[38] V. S. Murahari and T. Plötz, “On attention models for human activity recognition,” in *Proc. ACM Int. Symp. Wearable Comput.*, Oct. 2018, pp. 100–103.

[39] W. Gao, L. Zhang, Q. Teng, J. He, and H. Wu, “DanHAR: Dual attention network for multimodal human activity recognition using wearable sensors,” *Appl. Soft Comput.*, vol. 111, Nov. 2021, Art. no. 107728.

[40] S. P. Singh, M. K. Sharma, A. Lay-Ekuakille, D. Gangwar, and S. Gupta, “Deep ConvLSTM with self-attention for human activity decoding using wearable sensors,” *IEEE Sensors J.*, vol. 21, no. 6, pp. 8575–8582, Sep. 2021.

[41] M. A. Khatun et al., “Deep CNN-LSTM with self-attention model for human activity recognition using wearable sensor,” *IEEE J. Transl. Eng. Health Med.*, vol. 10, pp. 1–16, 2022.

[42] X. Yin, Z. Liu, D. Liu, and X. Ren, “A novel CNN-based bi-LSTM parallel model with attention mechanism for human activity recognition with noisy data,” *Sci. Rep.*, vol. 12, no. 1, pp. 1–11, May 2022.

[43] B. Khaertdinov, E. Ghaleb, and S. Asteriadis, “Contrastive self-supervised learning for sensor-based human activity recognition,” in *Proc. IEEE Int. Joint Conf. Biometrics (IJCB)*, Aug. 2021, pp. 1–8.

[44] Y. Shavit and I. Klein, “Boosting inertial-based human activity recognition with transformers,” *IEEE Access*, vol. 9, pp. 53540–53547, 2021.

[45] I. D. Luptáková, M. Kubovčík, and J. Pospíchal, “Wearable sensor-based human activity recognition with transformer model,” *Sensors*, vol. 22, no. 5, p. 1911, Mar. 2022.

[46] F. Moya Rueda, R. Grzeszick, G. Fink, S. Feldhorst, and M. T. Hompel, “Convolutional neural networks for human activity recognition using body-worn sensors,” *Informatics*, vol. 5, no. 2, p. 26, May 2018.

[47] J. Wang, Y. Chen, and Y. Gu, “A wearable-HAR oriented sensory data generation method based on spatio-temporal reinforced conditional GANs,” *Neurocomputing*, vol. 493, pp. 548–567, Jul. 2022.

[48] N. Ogawa, K. Kaji, and N. Kawaguchi, “Effects of number of subjects on activity recognition-findings from HASC2010corpus,” in *Proc. Int. Workshop Frontiers Activity Recognit. Using Pervasive Sens. (IWFAR)*, 2011, pp. 48–51.

[49] S. Yao et al., “SenseGAN: Enabling deep learning for Internet of Things with a semi-supervised framework,” *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 2, no. 3, pp. 1–21, Sep. 2018.

[50] A. Stisen et al., “Smart devices are different: Assessing and mitigating mobile sensing heterogeneities for activity recognition,” in *Proc. 13th ACM Conf. Embedded Netw. Sensor Syst.*, New York, NY, USA, Nov. 2015, pp. 127–140.

[51] A. Zahin, L. T. Tan, and R. Q. Hu, “Sensor-based human activity recognition for smart healthcare: A semi-supervised machine learning,” in *Artificial Intelligence for Communications and Networks*. Cham, Switzerland: Springer, 2019, pp. 450–472.

[52] X. Zhang, L. Yao, and F. Yuan, “Adversarial variational embedding for robust semi-supervised learning,” in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2019, pp. 139–147.

[53] A. Reiss and D. Stricker, “Introducing a new benchmarked dataset for activity monitoring,” in *Proc. 16th Int. Symp. Wearable Comput.*, Jun. 2012, pp. 108–109.

[54] L. Peng, L. Chen, Z. Ye, and Y. Zhang, “AROMA: A deep multi-task learning based simple and complex human activity recognition method using wearable sensors,” *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 2, no. 2, pp. 1–16, Jul. 2018.

[55] E. Sansano, R. Montoliu, and Ó. B. Fernández, “A study of deep neural networks for human activity recognition,” *Comput. Intell.*, vol. 36, no. 3, pp. 1113–1139, Aug. 2020.

[56] A. Odena, V. Dumoulin, and C. Olah, “Deconvolution and checkerboard artifacts,” *Distill*, vol. 1, no. 10, p. e3, Oct. 2016. [Online]. Available: http://distill.pub/2016/deconv-checkerboard

[57] K. He and J. Sun, “Convolutional neural networks at constrained time cost,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2014, pp. 5353–5360.

[58] R. Chavarriaga et al., “The opportunity challenge: A benchmark database for on-body sensor-based activity recognition,” *Pattern Recognit. Lett.*, vol. 34, no. 15, pp. 2033–2042, Nov. 2013.

[59] L. Gao, A. K. Bourke, and J. Nelson, “Evaluation of accelerometer based multi-sensor versus single-sensor activity recognition systems,” *Med. Eng. Phys.*, vol. 36, no. 6, pp. 779–785, Jun. 2014.

[60] A. Baldominos, P. Isasi, and Y. Saez, “Feature selection for physical activity recognition using genetic algorithms,” in *Proc. IEEE Congr. Evol. Comput. (CEC)*, Jun. 2017, pp. 2185–2192.

[61] Y.-C. Huang, C.-W. Yi, W.-C. Peng, H.-C. Lin, and C.-Y. Huang, “A study on multiple wearable sensors for activity recognition,” in *Proc. IEEE Conf. Depend. Sec. Comput.*, Aug. 2017, pp. 449–452.

[62] M. Zeng, T. Yu, X. Wang, L. T. Nguyen, O. J. Mengshoel, and I. Lane, “Semi-supervised convolutional neural networks for human activity recognition,” in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2017, pp. 522–529.

[63] R. Adaimi and E. Thomaz, “Leveraging active learning and conditional mutual information to minimize data annotation in human activity recognition,” *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 3, no. 3, pp. 1–23, Sep. 2019.

[64] Y. Jain, C. I. Tang, C. Min, F. Kawsar, and A. Mathur, “ColloSSL: Collaborative self-supervised learning for human activity recognition,” *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 6, no. 1, pp. 1–28, 2022.


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